11.2 Normal Modes

WORKSHEET

The question

Two particles X and Y, of equal mass m, lie on a smooth horizontal table and are connected by a light elastic spring of natural length a and modulus of elasticity λ. Two more springs, identical to the first, connect X to a point P on the table and Y to a point Q on the table. The distance between P and Q is $3a$.

Initially, the particles are held so that $XP = a$, $YQ = \frac{1}{2}a$, and $PXYQ$ is a straight line. The particles are then released.

At time t, the particle X is a distance $a + x$ from P and the particle Y is a distance $a + y$ from Q. Show that

$$m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -\frac{\lambda}{a}(2x + y)$$

and find a similar expression involving $\dfrac{\mathrm{d}^2y}{\mathrm{d}t^2}$. Deduce that

$$x - y = A\cos\omega t + B\sin\omega t$$

where A and B are constants to be determined and $ma\omega^2 = \lambda$. Find a similar expression for $x + y$.

Show that Y will never return to its initial position.

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself With X at distance $a + x$ from P and Y at distance $a + y$ from Q, how long is the middle spring XY? Which springs are stretched and which compressed when x and y are positive?

STAGE 2 Decoupling

Ask yourself Each equation involves both x and y. Which combinations of the two equations involve only one combination of x and y?

STAGE 4 Never returning

Ask yourself Write y in terms of t. For Y to be back at its initial position, $y = -\frac{1}{2}a$. What must both cosines be?

Try this next

1. **Energy.** Show that the total energy $\frac{1}{2}m(\dot{x}^2 + \dot{y}^2) + \frac{\lambda}{2a}\left(x^2 + y^2 + (x+y)^2\right)$ is constant, and that it splits into a sum of two terms, one for each normal mode, each of which is separately constant.

2. **Pure modes.** How should the particles be started so that the motion is periodic, with period $\frac{2\pi}{\omega}$? With period $\frac{2\pi}{\sqrt{3}\,\omega}$?

3. **A stiffer middle spring.** If the middle spring has modulus μ instead of λ, find the two normal-mode frequencies. For which values of μ/λ is every motion periodic?

4. **Three masses.** Three equal masses are connected in a line by four identical springs between two fixed points. Find the matrix and its eigenvalues $(2 - \sqrt{2},\, 2,\, 2 + \sqrt{2})$, and describe the three normal modes.

Further reading

- **A. P. French, *Vibrations and Waves*** (MIT Introductory Physics Series), Chapter 5, on coupled oscillators and normal modes.
- **Gilbert Strang, *Introduction to Linear Algebra*** (Wellesley-Cambridge), the chapter on eigenvalues, which includes springs and masses as a main example.
- **Steven Strogatz, *Nonlinear Dynamics and Chaos*** (CRC Press), Chapter 8, on quasi-

periodic motion on a torus.

The full guided solution, and what the question leads to at university, is in Section 11.2:
https://beyondstep.keithmcnulty.org/ch11/normal-modes.html